

STAGE 6 VERIFICATION REPORT: AI REPORT PLANNER

Verification Date: September 29, 2026 | Evaluation Suite: Automated PyTest v9.1.1 & Next.js Production Build | Overall Status: **VERIFIED & LOCKED**
(64/64 TESTS PASSED)

EXECUTIVE VERDICT: STAGE 6 PRODUCTION READY & FULLY LOCKED

Stage 6 of the ARM (AI Academic Report Assistant) platform has been thoroughly implemented, automated, verified, and locked. Stage 6 introduces the intelligent **AI Report Planner**, translating the authoritative Stage 4 Locked College Template, Stage 5 Project Information, and Stage 5 Evidence Locker metadata into a rigorous, fully structured ReportPlan. The planning engine operates under strict deterministic invariants: report sections must strictly mirror the locked college template without hallucinated or omitted sections; all facts are derived solely from student project data (with missing items explicitly flagged rather than fabricated); evidence is mapped strictly by registered locker metadata IDs; and post-generation plans are validated deterministically against the template schema before database persistence. Multi-tenant IDOR protection, stale plan invalidation, and an explicit student approval workflow are fully operational with 100% test pass rate across all stages.

1. Comprehensive Test Suite Execution Summary

A single end-to-end regression pass was executed across all testing suites. All 64 tests passed with zero failures and zero regressions against Stages 1–5.

Test Suite Module	Target Area	Tests	Result
test_stage6_report_planner.py	AI Report Planner, Gemini Integration, Validator, IDOR, Approval	7	PASS (7/7)
test_stage5_project_evidence.py	Project Info, Members, Evidence Locker, SHA-256, Readiness	8	PASS (8/8)
test_stage4_template_review_lock.py	Template Review, Canonical Normalization, Lock Immutability	10	PASS (10/10)
test_stage3_template_intelligence.py	Deterministic AST Parsing, Formatting Extraction, Schema Gen	9	PASS (9/9)
test_stage2_template_ingestion.py	DOCX/PDF Upload, Validation, Storage Fingerprinting, Queue	10	PASS (10/10)
test_stage1_verification.py	Database Foundation, RLS, Schema, Multi-Tenant Isolation	7	PASS (7/7)
test_foundation_services.py	Storage Abstraction, AI Provider Mock/Real, Error Handling	4	PASS (4/4)
test_health.py	Health Checks, v1 System Diagnostics, Root Endpoints	3	PASS (3/3)
test_arm_database_foundation.py	Core Connection Pooling & PostgREST Integrity	1	PASS (1/1)
test_api_endpoints.py	Baseline API Routers, Project CRUD, Legacy Dispatch	3	PASS (3/3)
test_template_parser.py	Standalone Parser Benchmark & Fallback Schemas	1	PASS (1/1)
test_document_assembler.py	Document Assembler DOCX Rendering Pipeline	1	PASS (1/1)
TOTAL SUITE VERIFICATION	All System Subsystems & Stages 1–6	64	PASS (64/64)

2. Stage 6 Dedicated Test Suite Details

Test Case Name	Target Behavioral Invariant	Result
test_plan_generation_requires_locked_template	Guarantees report planning is strictly rejected (HTTP 422 Unprocessable Content) if template is not locked.	PASS
test_plan_generation_and_template_structural_adherence	Executes live Gemini AI structured planning; verifies 1:1 section mapping, prompt injection immunity, and evidence tagging.	PASS
test_deterministic_validator_unit	Tests ReportPlanValidator enforcing section correspondence, blocking invented sections, and flagging missing facts.	PASS
test_multitenant_idor_protection	Verifies HTTP 403 Forbidden when Tenant B attempts to generate, read, inspect jobs, or approve Tenant A's plan.	PASS
test_plan_approval_workflow	Validates student approval endpoint transition from 'draft' to 'approved', locking plan for generation stages.	PASS
test_stale_plan_detection	Verifies that updating project info or evidence after plan generation flags 'is_stale=True' on subsequent plan retrieval.	PASS
test_prompt_injection_defense	Verifies that adversarial system instructions embedded in project facts/template are safely quarantined as passive text.	PASS

3. Architectural Invariants & Security Pillars

Pillar	Specification & Technical Implementation	Status
Template Invariant	Sections are 100% driven by the Stage 4 locked template. No invented sections, preserved hierarchy, ordering, page budget constraints, and mandatory flags.	ENFORCED
Ground Truth Anchor	All factual assertions are anchored strictly in Stage 5 project information. Where factual data is missing, the validator flags missing_information rather than fabricating facts.	ENFORCED
Evidence Mapping	Supporting evidence files (diagrams, benchmark datasets, survey forms) are mapped strictly by metadata ID from the Evidence Locker. Arbitrary file references are rejected.	ENFORCED
Research Flagging	Literature review and theoretical sections requiring academic backing are flagged with 'research_required=True' and topic tags without invoking unauthorized web agents or fake citations.	ENFORCED
Deterministic Post-AI Validator	Pydantic & rule-based ReportPlanValidator inspects raw LLM output, enforcing section exactness, recalculating summary metrics, and guaranteeing plan reliability.	ENFORCED
Multi-Tenant Isolation	PostgreSQL RLS policies + API layer project ownership checks ensure zero IDOR vulnerabilities across plans, generation jobs, and approvals.	ENFORCED
Stale Plan Detection	When project details or evidence are modified, subsequent plan reads flag 'is_stale=True', alerting students that the plan no longer reflects current project state.	ENFORCED
Scope Separation	Stage 6 handles PLANNING ONLY. Generation of final report prose, section writing, citation synthesis, and DOCX/PDF rendering are strictly separated and deferred to future stages.	ENFORCED

4. Database Schema & API Surface

Stage 6 introduces migration 20260929040000_arm_stage6_report_planner.sql creating the public.report_plans table with foreign keys, indexes, and RLS policies.

Entity / Endpoint	Type / Method	Description & Security Policy
public.report_plans	PostgreSQL Table	Stores plan JSON, approval status ('draft', 'approved', 'stale'), project FK, template FK, project_updated_at snapshot, and owner auth.uid().
POST /api/v1/projects/{id}/report-plan	REST Endpoint	Queues/executes AI report planning using locked template + project facts + evidence locker metadata. Ownership protected.
GET /api/v1/projects/{id}/report-plan	REST Endpoint	Retrieves current report plan for project. Evaluates stale condition against current project.updated_at timestamp.
POST /api/v1/projects/{id}/report-plan/approve	REST Endpoint	Transitions plan status to 'approved'. Authorizes the plan for future stage report generation.
GET /api/v1/projects/{id}/report-plan/jobs/{job_id}	REST Endpoint	Polls background planning job status ('running', 'completed', 'failed') in generation_jobs table.

5. Frontend Integration & Build Verification

The frontend user interface in apps/web was fully updated with the interactive ReportPlanner component allowing students to generate, review section readiness, view evidence linkages, identify missing data items, and approve plans.

Verification Check	Command	Observed Result	Status
TypeScript Compilation	npx tsc --noEmit	0 errors across all Next.js 15 pages and components	PASS
ESLint Quality Check	npm run lint	0 warnings, 0 errors, strict code formatting verified	PASS
Production Build	npm run build	Optimized build completed; all 26 static routes successfully generated	PASS

STAGE 6 FORMAL CERTIFICATION & LOCK ATTESTATION

All verification criteria for Stage 6 (AI Report Planner) have been satisfied. The planning pipeline is deterministic, securely sandboxed, and anchors all future report writing to the locked template and student-provided evidence. Stage 6 is officially **LOCKED**. Per system constraints, Stage 7 has not been started.